ASCOMP 2026 POSTER

Min-Marginal Power Indices for Multi-Objective Feature Selection

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01 PROBLEM & METHOD

Average importance can hide a harmful coalition:
adding a feature can turn an acceptable model into an unacceptable one.

Acceptable
$v(S) = 1$

+ feature i

marginal = -1

Unacceptable
$v(S \cup \{i\}) = 0$

THE WORST-CASE MARGINAL

$$v_\alpha(S) = 1[U(M_S) \geq \alpha\, U_{full}]$$

$$\mathbf{MinPI_i = \min [v_\alpha(S \cup \{i\}) - v_\alpha(S)]}$$

over $S \subseteq N \setminus \{i\}$; N = analyzed features

Validate before removal

H: features with an observed -1 marginal on validation data.

$$\mathbf{U_{val}(N \setminus H) \geq U_{val}(N) - \varepsilon}$$

ε = 0.005 (classification); 0.01 (regression).

Pass: remove H. Fail: remove nothing. Models fit on training data.

SCAN ME

Scan to read the full paper

A witnessed -1 certifies harm in one coalition.
Sampling can still miss harmful witnesses.

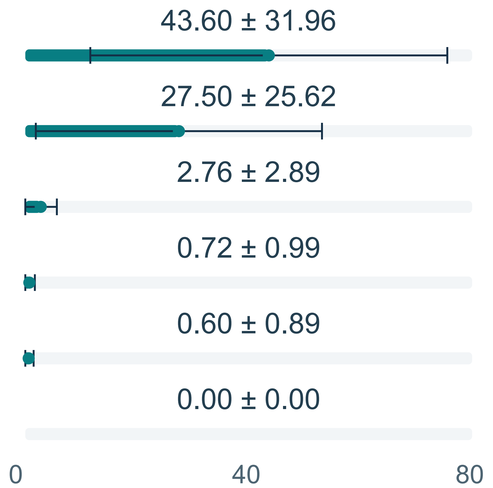
hlnks.co/63671101

02 EXPERIMENTS & RESULTS

6 datasets × 5 seeds | XGBoost | α = 0.95
64/16/20% train/validation/test; ≤100 analyzed features.

Five-seed mean ± SD; removal of analyzed features.
ΔU = held-out utility - full-model utility.

DATASET	REMOVED (%)	HELD-OUT ΔU
Madelon	43.60 ± 31.96	+0.00469 ± 0.00497
Abalone	27.50 ± 25.62	-0.00565 ± 0.01077
Credit-Card-Fraud	2.76 ± 2.89	-0.00078 ± 0.00335
Spambase	0.72 ± 0.99	-0.00003 ± 0.00007
Gisette	0.60 ± 0.89	-0.00011 ± 0.00016
ISOLET	0.00 ± 0.00	0.00000 ± 0.00000



12.53%
mean feature removal

-0.000313
mean held-out ΔU

NSGA-II: +0.01584 mean held-out ΔHV (800 evaluations).
Holm p = 0.4375: not significant; mainly driven by Madelon.

The gate does not guarantee held-out non-inferiority.
No overall superiority or speedup is established.